

Academic Excellence Through Quality Education
KATHMANDU MODEL SECONDARY SCHOOL
FIRST TERM EXAMINATION 2082

Class: XII 'Science'

SET-A

Time: 25 min

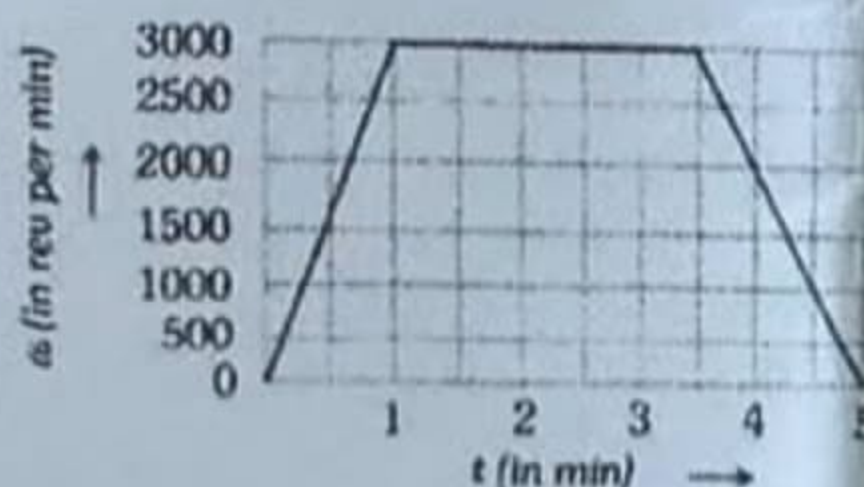
Subject: Physics [0031]

Re-write the best option among the four options on your answer sheet.

Attempt all the questions.

GROUP 'A' MCQ [11×1=11]

1. As a part of maintenance inspection, the compressor of a jet engine is made to spin according to the graph as shown. The number of revolution made by the compressor during the test is:



- a) 9000 b) 16570 c) 12750 d) 11250
2. Two particles execute SHM of same frequency and amplitude along the same straight line. They pass one another while going in opposite directions. Each time their amplitude is half of their amplitude. The phase difference between them is:
- a) 30° b) 60° c) 90° d) 120°
3. A tangential force F acts along the rim of a ring of radius R and displaces the ring through an angle θ . The work done by the force is:
- a) $FR^2\theta$ b) $FR\theta$ c) $\frac{FR}{\theta}$ d) $\frac{FR}{2}\theta$
4. A watch based on oscillating spring gives correct time on earth. If it is taken to the moon. Then it
- a) Becomes fast b) becomes slow
 c) remains unaffected d) stops
5. The frequency of a sound wave is f and its velocity is v . If frequency is increased to $4f$, the velocity of the wave will be
- a) v b) $2v$ c) $4v$ d) $v/4$

6. E_n and L_n denote the magnitude of total energy and angular momentum of an electron in n^{th} orbit of a Bohr atom, then:

a) $E_n \propto L_n$ b) $E_n \propto \frac{1}{L_n}$ c) $E_n \propto L_n^2$ d) $E_n \propto \frac{1}{L_n^2}$

7. Which of the following property of wave is independent of the other?

- a) velocity b) frequency
c) amplitude d) wavelength

8. When a long steel pipe is tapped at one end, a listener hears two sounds:

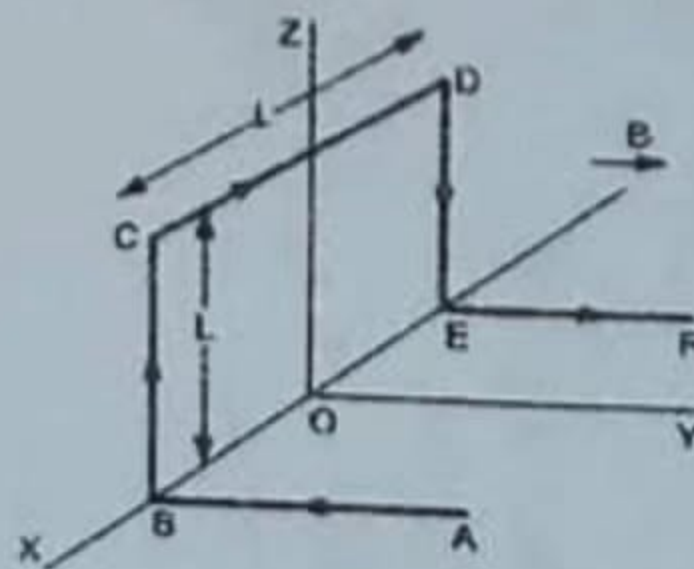
- a) of the same intensity, at same time
b) of the same intensity, one after another
c) less intense first, more intense later
d) more intense first and less intense later

9. An empty vessel is getting filled with water, then its frequency will

- a) go on increasing b) go on decreasing
c) remains unchanged d) insufficient information

10. A wire ABCDEF is bent in the form as shown in the figure. The wire carries a current I and is placed in a uniform magnetic field of induction B parallel to positive Y-axis. If each side is of length L , the force experienced by the wire will be:

- a) IBL along the positive Z-direction
b) IBL along the negative Z-direction
c) $2IBL$ along the positive Z-direction
d) $2IBL$ along the negative Z-direction



11. An electron is projected in the direction parallel to both electric and magnetic field, electron will

- a) Magnitude of velocity goes on increasing
b) Magnitude of velocity goes on decreasing
c) At first magnitude of velocity decreases and then increases
d) There will be no change in magnitude of velocity.

FIRST TERM EXAMINATION 2082

Class: XII 'Science'

SET-B

Time: 25 min

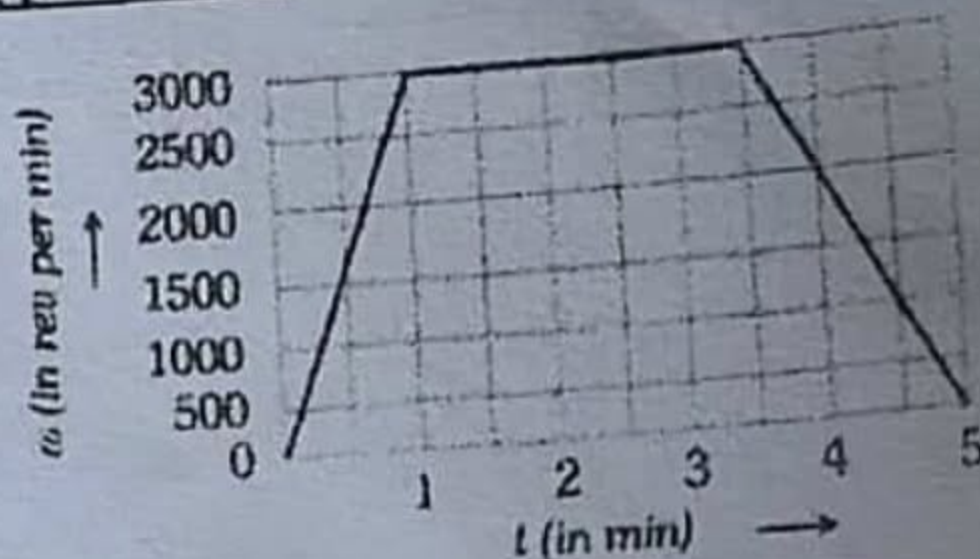
Subject: Physics [0031]

Re-write the best option among the four options on your answer sheet.

Attempt all the questions.

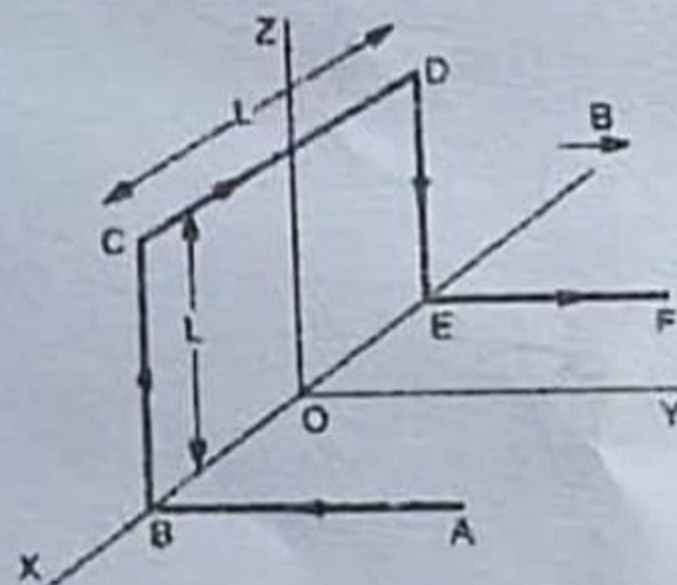
GROUP 'A' MCQ [11×1=11]

1. As a part of maintenance inspection, the compressor of a jet engine is made to spin according to the graph as shown. The number of revolutions made by the compressor during the test is:



- a) 11250 ☒ b) 12750 c) 16570 ☒ d) 9000
2. Two particles execute SHM of same frequency and amplitude along the same straight line. They pass one another while going in opposite directions. Each time their ~~amplitude~~ ^{displacement} is one-third of their ~~amplitude~~ ^{amplitude}. The phase difference between them is: ~~displacement~~
- a) 141° b) 35.25° c) 70.5° d) 105.75°
3. A torque τ is applied to a uniform disc of radius R and it rotates through an angle θ . The work done by the torque is::
- a) $\tau R^2 \theta$ b) $\tau R \theta$ c) $\tau \theta$ d) $\frac{\tau \theta}{2}$
4. A mass-spring system oscillates with a certain period on Earth. If the spring constant is doubled, the time period of oscillation will:
- a) Double b) Decrease by a factor of $\sqrt{2}$
- c) Increase by a factor of $\sqrt{2}$ d) Halve
5. The wavelength of a sound wave is λ and its velocity is v . If wavelength is increased to 4λ , the velocity of the wave will be
- a) v b) $2v$ c) $4v$ d) $v/4$

6. In a hydrogen-like atom, the radius of the n^{th} orbit is r_n and the speed of the electron in that orbit is v_n . If T_n and V_n denote the kinetic and potential energies of the electron in the n^{th} orbit, then:
- a) $T_n \propto r_n$ b) $T_n \propto \frac{1}{r_n}$ c) $V_n \propto r_n$ d) $V_n \propto \frac{1}{r_n^2}$
7. Which of the following property of wave is independent of the other?
- a) amplitude b) wavelength
c) velocity d) frequency wavelength
8. When a long steel pipe is tapped at one end, a listener hears two sounds:
- a) of the same intensity, at same time
b) of the same intensity, one after another
c) less intense first, more intense later
d) more intense first and less intense later
9. An air column closed at one end is set into vibration and emits a fundamental note of 550 cycles per second. If the speed of sound in air is 330 m/s, the wavelength of the sound in the surrounding air is:
- a) 30 cm b) 60 cm c) 15 cm d) 45 cm
10. A wire ABCDEF is bent in the form as shown in the figure. The wire carries a current I and is placed in a uniform magnetic field of induction B parallel to positive Y-axis. If each side is of length L , the force experienced by the wire will be:
- a) $2IBL$ along the positive Z-direction
b) $2IBL$ along the negative Z-direction
c) IBL along the positive Z-direction
d) IBL along the negative Z-direction
11. A proton is projected in the direction parallel to both electric and magnetic field, proton will
- a) magnitude of velocity goes on decreasing
b) magnitude of velocity goes on increasing
c) There will be no change in magnitude of velocity.
d) At first magnitude of velocity decreases and then increases



FIRST TERM EXAMINATION 2082

Class: XII 'Science'

F.M.: 75

Time: 3 hrs.

Subject: Physics [1021]

P.M.: 30

Candidates are required to give their answer in their own words as far as practicable.

The figure in the margin indicated full marks:

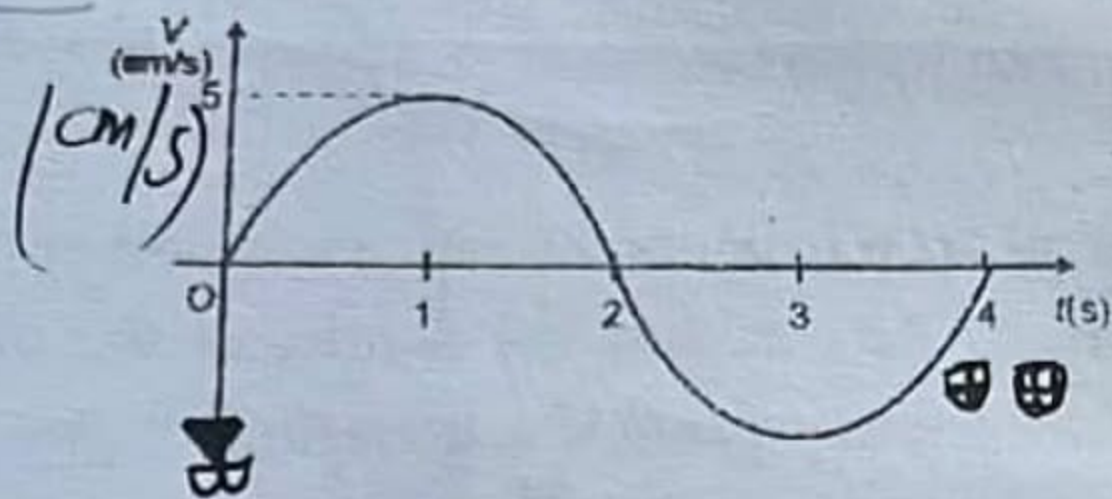
Attempt all the questions.

GROUP 'B' [8×5=40]

Short answer questions:

12. a) Does the moment of inertia of a body become zero as soon as it stops rotating? Explain. [1]
b) Is the radius of gyration of a body a constant quantity? If not then, on which factors does it depend? [1]
c) Derive an expression for moment of inertia of a thin uniform rod about an axis passing through one end and perpendicular to its length. What will be the result if the axis passes through the center? [3]
13. a) State and prove the principle of conservation of angular momentum with suitable example. [2]
b) Why the middle seat is more comfortable while travelling in a bus in a bumpy road? [1]
c) A flywheel of moment of inertia 0.32 Kg m^2 is rotated steadily at 120 rad s^{-1} by a 50 W electric motor (i) Find the kinetic energy and angular momentum of the flywheel. (ii) Calculate the value of the frictional couple opposing the rotation. [2]
14. a) Write the characteristics of simple harmonic motion. [1]
b) Show mathematically and graphically the total energy of a particle executing SHM is constant. [3]
c) A particle executes SHM between $x = \pm 0.1 \text{ m}$. When its displacement is $x = 0.05 \text{ m}$, its speed is 0.6 m/s . Find the total energy if $m = 0.2 \text{ kg}$. [1]
15. a) What is the principle of superposition of waves? Discuss the result of superposing of two waves of equal amplitude and same frequency travelling in opposite direction. [3]

- b) A certain transverse sinusoidal wave of wavelength 20 cm is moving in the positive x direction. The transverse velocity of the particle at $x = 0$ as a function of time is shown. Find
- i) the amplitude of the wave and ii) write the displacement equation that describes the wave. [2]



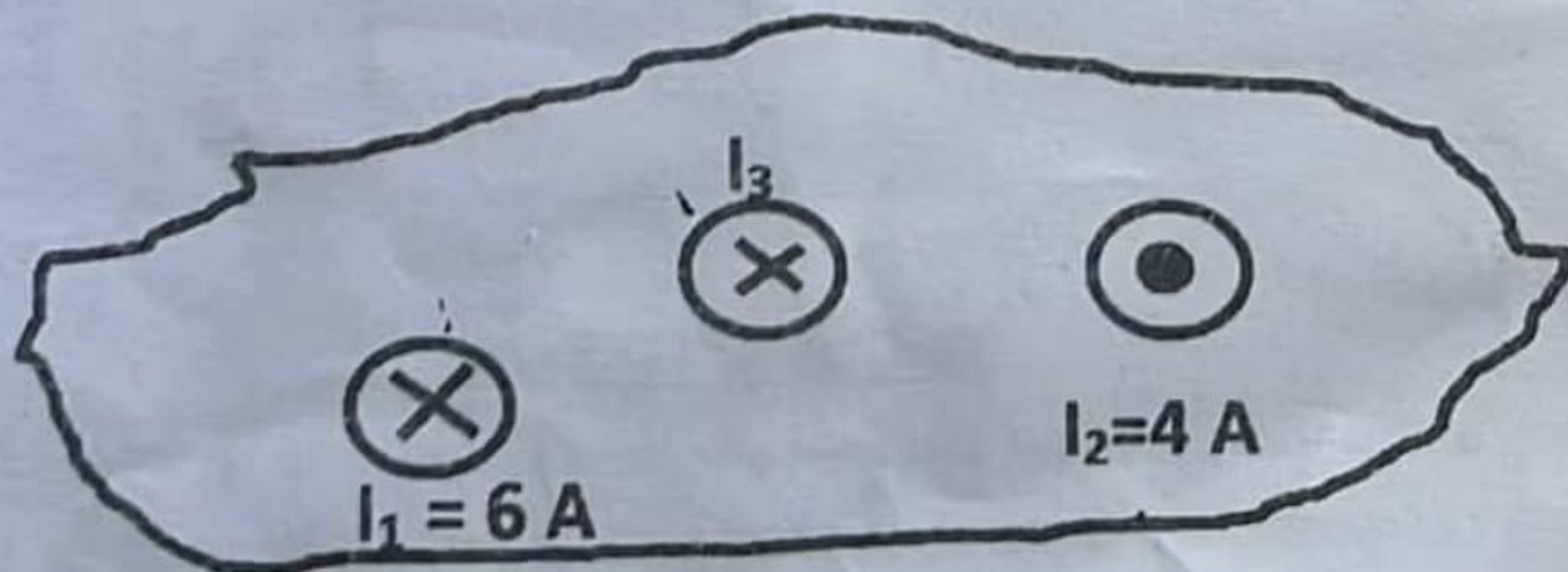
16. a) Write the expression for the magnitude of the magnetic Lorentz force, stating the physical quantities involved. Also, explain the Fleming's left hand rule used to determine the direction of the force. [2]

- b) A proton moves in a uniform magnetic field of strength 0.500 T directed along the x-axis. At initial time, $t = 0$ s, the proton has velocity, $\vec{v} = (1.95 \times 10^5 \hat{i} + 2.00 \times 10^5 \hat{k})$ m/s. Find

- (i) At initial time, what is the acceleration of the proton?
- (ii) Is the path circular or helical? If helical, calculate the radius of helical trajectory and also calculate the pitch of the helix. [3]

17. a) What is toroid? Use Ampere's circuital law to find out the magnetic field inside a toroid carrying current. [3]

- b) The value of line integral of $\vec{B}$ around the closed path as shown in figure is 3.77×10^{-6} Tm. Find the value of I_3 ? [2]

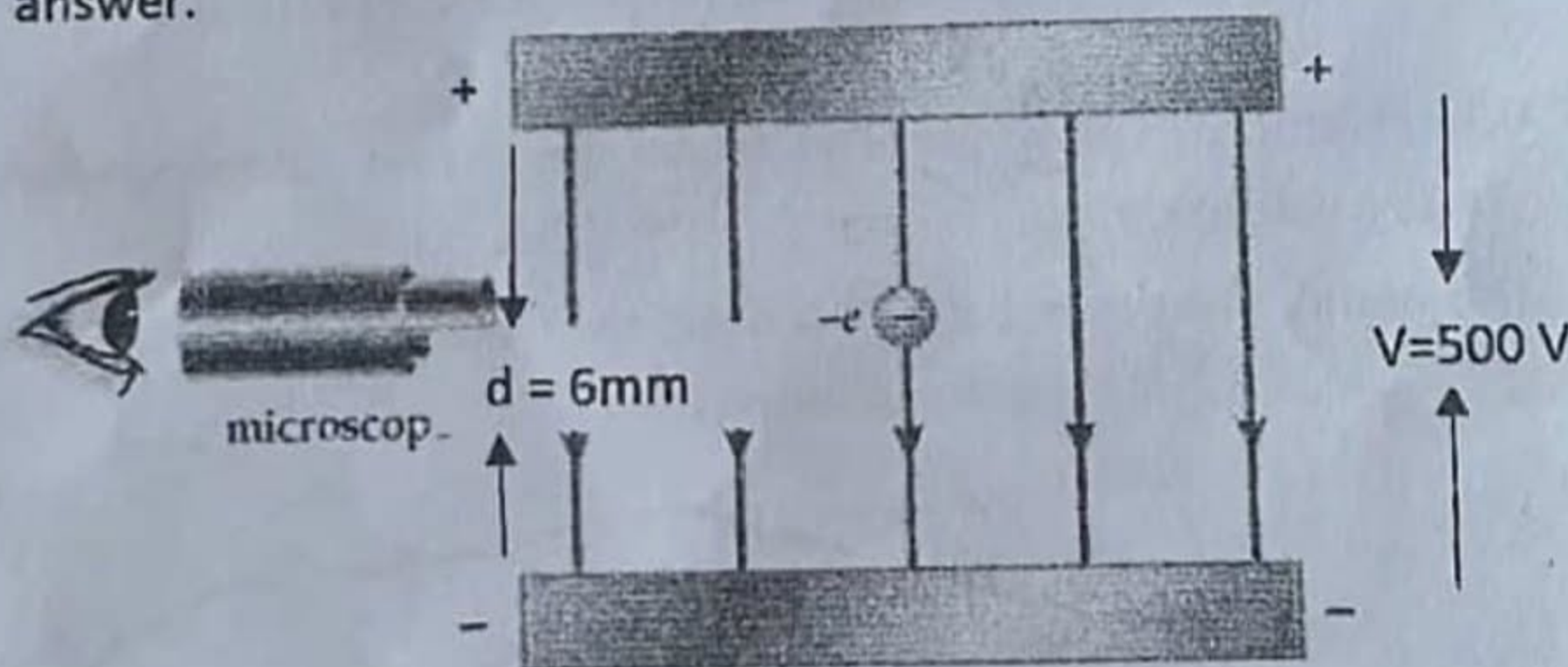


18. a) State the postulates of Bohr's theory of hydrogen atom. [2]

- b) Show that the speed of an electron in the first excited atom of hydrogen atom is $1/274$ times the speed of light. [2]
- c) Explain why the electron moves relatively faster in the inner orbit as compared to the outer orbit? [1]

19. Millikan's oil drop experiment was replicated in the laboratory. The droplet radius was 7×10^{-7} m, and the density of the oil was 900 kg m^{-3} . When a potential difference of 500 V was applied to the plates, the oil droplet was observed to be stationary.

- a) Calculate the charge on the oil drop. [2]
- b) Comment on the most likely measurement to be responsible for the discrepancy between the calculated charge and the accepted value of the charge on the electron. [1]
- c) If the potential difference is raised above 500 V what will happen to the oil drop? [1]
- d) If the polarity of the plates is reversed while keeping the potential difference at 500 V, what will happen to the oil drop? Explain your answer. [1]



GROUP 'C' [3×8=24]

20. During a thunderstorm, a student observes lightning first and hears thunder after 3 seconds. The temperature of air is 25°C . The speed of sound in air at 0°C is 332 m/s .

- a) Derive the expression for the speed of sound in air starting from the assumptions of Laplace's correction in terms of temperature. [3]
- b) Calculate the distance of the lightning strike from the observer. [2]

- c) Discuss how humidity affects the speed of sound in air. [1]
 d) "What do you mean by harmonics? Is the statement 'All harmonics are overtones, but all overtones are not harmonics' correct? If not, explain." [1]



21. a) A current element produces a magnetic field even though it is electrically neutral. Explain how this is possible. [1]
 b) State Biot-Savart law. [1]
 c) Use Biot-Savart law to find out the magnetic field at a point due to a long straight conductor carrying current. [3]
 b) A closely wound, circular coil with radius 2.40 cm has 800 turns. (i) What must be the current in the coil be if the magnetic field at the center of the coil is 0.0580 T? ii) At what distance from the center of the coil on the axes of the coil, the magnetic field is $1/8$ times that at the center of the coil? [3]

22. a) The work function of a metal is 6.4×10^{-19} . Explain what is meant by the term work function. Light with frequency of 1.2×10^{15} Hz is shone onto the metal surface. Find whether or not the photons of this light will cause photoelectric effect to take place. [2]
 c) Draw a graph showing the variation of stopping potential with frequency of radiation incident on a metal plate. How can the value of Planck's constant be determined from the graph? [3]
 d) From the following data, plot a graph for the photo-electric emission and find the value of threshold frequency, work function and Planck's constant? [3]

Frequency of incident light, f	3×10^{14} Hz	4×10^{14} Hz	5×10^{14} Hz	6×10^{14} Hz	7×10^{14} Hz
Maximum K.E of emitted photo electrons, E	0.1 eV	0.5 eV	0.9 eV	1.3 eV	1.7 eV



Best of Luck